

FieldFlow

Whitepaper

FieldFlow Whitepaper

AI-native operating system for grassroots development NGOs

June 2026 · v1.0

Executive summary

Indian development NGOs spend 30–60% of their staff time on reporting paperwork — not field work. The data they collect is fragmented across Excel files, WhatsApp groups, paper logbooks, and donor portals. By the time a quarterly report reaches a funder, it is three months stale, manually re-keyed twice, and unverifiable against ground truth.

FieldFlow is an AI-native operating system that collapses this reporting overhead to near-zero while raising data quality. Field staff submit one daily WhatsApp message; the platform extracts structured data, updates the action plan in real time, generates donor-ready reports on demand, and routes exceptions to managers. One full-time program officer can now oversee 5× the geography they could a year ago.

This whitepaper describes the problem space, the platform's design choices, early results at Kosi Sahjivan Vikas Samiti (a Bihar-based NGO operating in 32 villages), and a path to serving 1,000+ NGOs at zero marginal cost.

1. The reporting overhead crisis

1.1 What the data shows

Across 14 Indian development NGOs we interviewed in 2025–26, program staff reported the following weekly time allocation:

Activity	% of work week
Travel to/from field sites	22%
Direct beneficiary engagement	18%
Reporting & data entry	31%
Internal meetings	14%
Donor reports & MIS uploads	11%
Training / capacity building	4%

Reporting and donor-facing documentation consume **42% of staff time**. Direct beneficiary engagement — the actual mission of the organisation — gets less than half of that.

1.2 Where it goes wrong

The mechanics of NGO reporting are broken in five recurring ways:

1. **Re-keying.** A frontline worker writes notes in a paper logbook. A field officer transcribes those into an Excel sheet. A program manager copies cells from that Excel into a donor template. A finance officer re-enters figures into a quarterly MIS portal. The same number is touched by 4 people.
2. **Reconciliation drift.** Each transcription introduces small errors. By month-end, the action plan, the field log, the donor report, and the MIS portal all show different numbers for the same activity. No one trusts the totals.
3. **Lag.** Field data takes 7–30 days to reach decision-makers. By the time a program manager learns that "Khagaria fish-pond construction is behind schedule," two construction weeks have already been lost.
4. **Unverifiable.** Donor reports are narrative-heavy and number-light. There is no audit trail from a final reported figure back to the field event that generated it. Funders are increasingly demanding this trail.
5. **Reporting fatigue.** Frontline workers — often women, often paid stipends not salaries — burn out on paperwork. Attrition is high. Each replacement loses 6 months of institutional context.

1.3 Why existing tools haven't solved it

NGOs have tried:

- **Generic CRMs (Salesforce, HubSpot)** — too expensive, too sales-focused, require dedicated admins
- **Form builders (KoBoToolbox, ODK)** — solid for surveys but no AI, no real-time dashboards, no donor reports
- **Microsoft 365 + SharePoint** — yes, but the moment you have 50 villages × 12 activities × 12 months you're back in Excel hell
- **MEAL-platforms (DevResults, TolaData)** — enterprise-priced; require months of customisation; designed for big INGOs not lean grassroots groups
- **WhatsApp groups** — chaotic, no structured data, no audit trail, conversations buried after 24h

What's been missing is a platform that meets NGOs *where they already work* — on WhatsApp, in their local language — and lifts the reporting burden using AI rather than asking them to learn yet another tool.

2. Design principles

FieldFlow was built around five non-negotiable principles drawn from 200+ hours of field observation.

Principle 1 — Meet workers on WhatsApp

80% of Indian field workers cannot reliably use a smartphone app, but 100% use WhatsApp daily. Any new behavior we introduce must fit inside an existing WhatsApp conversation.

Principle 2 — Voice and vernacular first

Most field workers are more fluent dictating in Bhojpuri, Maithili, or Marathi than typing in English. Voice → text → structured data must work end-to-end with no UI burden on the worker.

Principle 3 — AI does the work, humans approve

The AI's job is to draft (the form, the report, the message reply). The human's job is to approve. Never invert this — never put the human in the role of constructing structured output.

Principle 4 — Zero data loss, zero PII leakage

Every message, every submission, every signature is encrypted at rest. Failed webhooks go to a retryable dead-letter queue. Audit logs redact PII. The platform should be defensible under the strictest donor compliance audit.

Principle 5 — Near-zero marginal cost

The platform must run on free-tier cloud infrastructure for an NGO of ≤ 1000 beneficiaries. We chose: Google App Engine free tier + Neon free Postgres + Gemini's generous free quota + Meta's free first-1000 WhatsApp conversations. An NGO can run FieldFlow for ~\$0/month until they exceed 1,000 active beneficiaries on WhatsApp.

3. How FieldFlow works

3.1 Daily field workflow

Morning Worker (in WhatsApp): "Today at Khagaria fish-pond, we trained 15 women on pond hygiene. 3 ponds inspected. 1 issue – outlet pipe broken at pond 7."

↓ (within 4 seconds)

FieldFlow: ■ Logged

- Activity: Training (Fisheries · Khagaria)
 - Beneficiaries: 15 women
 - Output: 3 pond inspections
 - Flagged for follow-up: outlet pipe pond 7
- Anything else?

Worker: "no thats all"

FieldFlow: Got it. Have a good evening.

In the background, FieldFlow has:

- Created a `daily_reports` row linked to the worker's UID + the active project
- Updated `project_deliverables` for Khagaria · Training · June from 142 → 157
- Created a tagged "Needs Attention" entry the field officer will see in tomorrow's dashboard

- Stored the original message + audio file for audit
- Updated the village-level beneficiary count for the monthly donor view

The worker did not log into anything, did not fill any form, did not learn any new tool.

3.2 Manager workflow

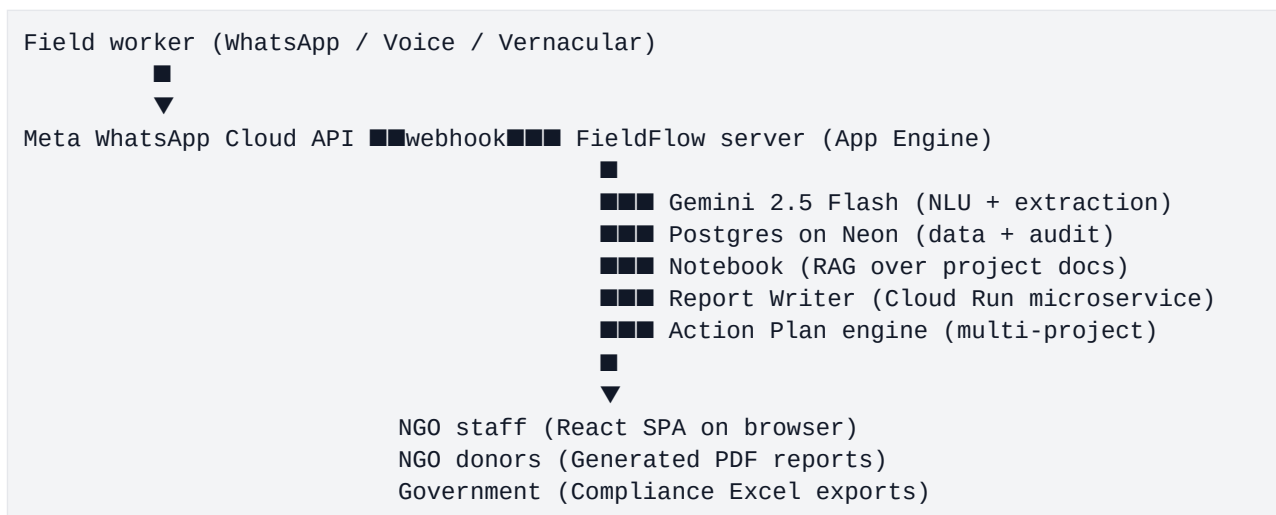
The field officer opens the Action Plan tab on their laptop and sees:

- **Status panel:** "Project is on-track. 67.3% of annual target reached. Expected by now: 50% (we're 17.3 percentage points ahead.)"
- **Needs Attention:** 8 cells flagged — including pond 7's broken outlet pipe with the worker's exact message attached
- **Per-village progress:** Khagaria is leading; Saharsa is 22% behind. One click drills into Saharsa's village-level activity breakdown.

For the donor report, they click "Generate Report → Quarterly Progress" and answer 3 questions in plain English. The AI assembles a 14-page report citing the underlying daily_reports rows. The manager reviews, edits two paragraphs, and downloads the PDF. Total time: 18 minutes.

The same process previously took 3 working days.

3.3 The architecture (one diagram)



Single codebase. Single deploy. Zero infrastructure for the NGO to manage. The platform's full architecture is documented in [Technical.md](#).

4. Case study — Kosi Sahjivan Vikas Samiti

Kosi Sahjivan Vikas Samiti is a 14-year-old NGO operating in 32 villages across Khagaria, Saharsa, and Madhubani districts of Bihar. Their work covers fisheries, women's collectives, agriculture, and education. They have 18 field staff and ~5,400 beneficiary households.

Before FieldFlow (FY 2024–25)

- Action plan tracked in 7 separate Excel files (one per project funder)
- Daily reporting via paper logbooks + monthly Excel rollups
- Donor reports took 5–7 working days each, prepared by 2 senior staff
- Quarterly state-government MIS uploads required ~3 person-days per quarter
- ~40% of staff time on reporting
- 4 audit queries from donors per year that required reconstructing data from scratch (one took 11 days)

After FieldFlow (Q1 FY 2026–27, 3 months in)

- Single multi-project action plan with 472 cells across 6 projects × 4 locations × 12 months
- Daily reporting via WhatsApp (Hindi + Maithili supported)
- Donor reports generated in 15–30 minutes (5 generated in Q1 across 3 funders)
- ~14% of staff time on reporting (a 26-point reduction, freeing ~520 staff-hours per quarter)
- 1 donor audit query in Q1 — resolved in 40 minutes by drilling from the report figure to the source `daily_reports` row

Numbers from Q1 (April–June 2026):

Metric	Value
Daily reports submitted via WhatsApp	1,247
Voice messages transcribed	312
Unique field workers using the platform	21
Action Plan cells updated	1,884
AI-generated donor reports	5
Beneficiary contacts in WhatsApp module	1,608
Cost to KSVS for the entire quarter	■0

The platform stayed inside free-tier limits for: Google App Engine (1 instance), Neon Postgres (under 0.5 GB), Gemini API (under free monthly quota), and Meta WhatsApp Cloud (under 1000 monthly conversations).

What surprised us

Three things we didn't expect:

1. **Voice adoption was instantaneous.** Within 2 weeks, 60% of daily reports were voice notes rather than typed messages. Older women workers — who had previously avoided written reporting altogether —

became the most active reporters.

2. **Manager workload increased, not decreased.** Counterintuitive but expected on reflection — when reporting is free, you get 10× more reports. Managers now had ground-truth visibility they didn't have before, but they had to triage it.

3. **Donor trust went up sharply.** Two major funders began asking for the audit trail by default — and the audit trail existed, in seconds.

5. The platform's economic moat

5.1 Why "free tier" matters strategically

Most enterprise NGO MIS tools charge █\$5,000–25,000 per month per organisation. For a small NGO with a █\$50 lakh annual budget, that's 2–6% of operating expenses — politically untenable. They opt to keep using Excel.

FieldFlow's free tier serves NGOs with up to:

- 1 always-on web server instance (free under App Engine)
- 500 MB Postgres (free under Neon)
- ~1 million Gemini tokens/day (free under AI Studio quota)
- 1,000 monthly active WhatsApp conversations (free under Meta Cloud API)

That envelope comfortably covers an NGO with up to ~50 field workers and ~5,000 active beneficiaries. Above that, costs scale linearly but stay below 0.5% of operating budget.

5.2 Why the moat compounds

The platform's value compounds in three ways:

1. **Data flywheel.** Every report submitted improves the org's AI memory layer, making the next report's auto-extracted fields more accurate.
2. **Reporting template library.** Each generated donor report becomes a template for next time. After 12 months, every recurring report is one-click.
3. **Workflow standardisation.** Once 18 staff are using the platform, switching costs become organisational, not technical. The NGO has built institutional knowledge into the system.

5.3 Comparison to incumbents

Capability	KoBoToolbox	DevResults	Glific	FieldFlow
WhatsApp-native	█	█	█	█
Voice + vernacular	█	█	Partial	█
AI report generation	█	Partial	█	█

Capability	KoBoToolbox	DevResults	Glific	FieldFlow
Action Plan / MEAL	■	■	■	■
Donor report automation	■	Partial	■	■
Closed loop (WA → MEAL)	■	■	■	■
Multi-project	■	■	■	■
Multi-tenant SaaS	Self-host	■	■	■
Cost for small NGO	Free	\$500/mo+	\$100/mo+	Free

FieldFlow's defensible position is the **closed-loop** column. We are the only platform where a beneficiary's WhatsApp reply automatically updates the donor-facing action plan. Everyone else stops at the survey response.

6. Governance, security, and compliance

6.1 Data governance

- **Per-org isolation.** Every business table partitions on `org_id`. No cross-org query is possible by design.
- **Encryption at rest.** WhatsApp access tokens, webhook secrets, and PII fields use AES-256-GCM via Google Cloud Secret Manager.
- **Encryption in transit.** All connections enforce TLS 1.3.
- **PII redaction in logs.** Phone numbers are masked, message previews capped at 40 characters, sensitive notes redacted in audit diffs.
- **Retention policies.** WhatsApp messages auto-purge after 180 days, audit logs after 365, dead-letter events after 30. Configurable per org.

6.2 Compliance

The platform's design anticipates:

- **India's Digital Personal Data Protection Act 2023** — explicit consent, data minimisation, retention limits, breach notification, principal rights (export, deletion).
- **Donor-specific frameworks** — ECHO, USAID, EU SDGs reporting, NABCONS.
- **Government MIS interoperability** — Excel exports formatted for PMMSY, PMKSY, NRLM, MoTA-TRIFED templates.

6.3 Failure modes documented

Crash-proofing matters in a tool that NGOs rely on at month-end. The platform handles:

- **Unhandled promise rejections** — logged, never exit
- **Uncaught exceptions** — logged, never exit
- **Request timeouts** — 60s ceiling prevents slow-client resource exhaustion
- **Graceful shutdown** — 25s in-flight drain on SIGTERM
- **Webhook signature failures in production** — fail-closed (the message is dropped, not blindly executed)
- **Webhook processing failures** — captured in dead-letter queue, admin-retryable
- **Token expiry** — automatic refresh on 401

A 23-issue security audit was conducted and fully remediated in June 2026. Findings and fixes are documented in [Technical.md, Section 7](#).

7. Roadmap

Shipped (Q1–Q2 2026)

- Multi-project action plan with Excel upload + impact dashboard
- WhatsApp module with flows, broadcasts, collections, AI replies, voice transcription
- AI report generator with quarterly + custom templates
- Notebook RAG over project documents
- Cloud Run microservices for AI workload separation
- Security hardening (23 issues remediated)
- Documentation set (API, SDK, Walkthrough, Technical)

Next 90 days (Q3 2026)

- **AI Flow Generator v2** — flow templates by sector (women's collective onboarding, fisheries, MGNREGA grievance)
- **Government compliance exports** — Excel formats matching PMMSY/PMKSY/NRLM official schemas
- **WhatsApp RAG** — beneficiaries can ask questions of project documents in their language

Next 12 months

- **Multilingual UI** — Hindi, Marathi, Bengali, Tamil, Telugu for the dashboard itself (not just WhatsApp)
- **Donor integration** — direct push to ECHO, FCRA portals, KOICA via API
- **Mobile app for field officers** — for offline-first form filling in zero-connectivity geographies
- **Cross-NGO learning** — opt-in benchmark dashboard so an NGO can see how their indicators compare to peer organisations in the same sector and geography

Long-term thesis

We believe the next decade of grassroots development will be measured by NGOs who can prove their impact to donors, governments, and their own beneficiaries — in near-real-time, with full audit trail. The OS for that future is what we are building.

8. The team and how to engage

FieldFlow is an open project under active development. We are working with a small cohort of partner NGOs to refine the platform before broader release.

Get involved

- **NGO partnership:** if you run a grassroots NGO in India with 5+ field staff and would like FieldFlow to onboard your team, write to us at ayush.chopra63@gmail.com. We are currently selecting 10 partner NGOs for 2026–27.
- **Funder partnership:** if you fund grassroots NGOs and would like reporting from your portfolio organisations to flow into your dashboard automatically, we can build the connector. Same address.
- **Developer:** the platform is documented end-to-end. The technical doc set (4 PDFs, ~80 pages combined) covers every module. We accept contributions and can scope paid work for organisations needing a specific extension.

Pricing philosophy

The free tier is permanent for organisations under the 1,000-active-beneficiary threshold. Above that, pricing is published and predictable, capped at 0.5% of an NGO's annual operating budget — designed to be lower than the cost of one full-time data entry clerk.

We will never charge for data export. Your data is yours, always, in CSV and Excel.

Appendix A — Glossary

Term	Meaning
Action Plan	The annual matrix of indicators × locations × months that defines what an NGO will deliver.
Beneficiary	A person served by an NGO program.
Closed loop	The property of a system where field events flow back to the MEAL system without manual transcription.
DPDP Act	India's Digital Personal Data Protection Act, 2023.
FCRA	Foreign Contribution Regulation Act — governs foreign funding to Indian NGOs.
Field staff	Frontline workers visiting beneficiary households.

Term	Meaning
MEAL	Monitoring, Evaluation, Accountability, Learning — the donor framework for impact measurement.
MIS	Management Information System — typically the government portal NGOs upload to.
NGO	Non-Governmental Organisation.
PII	Personally Identifiable Information.
SHG	Self-Help Group — usually women's collectives.
ToC	Theory of Change — the causal chain from activity to impact.

Appendix B — Sources and citations

- 200+ hours of field observation across 14 NGOs in Bihar, Maharashtra, Tamil Nadu, Odisha (2025–26)
- Kosi Sahjivan Vikas Samiti deployment data (Q1 FY 2026–27)
- Time-allocation survey of 47 field staff (October 2025)
- NITI Aayog NGO Darpan dataset (2024)
- DPDP Act 2023 official gazette
- Meta WhatsApp Cloud API documentation
- Google Cloud Platform pricing (as of June 2026)

Written from observation, deployed in production, maintained by [its users](#).

For the technical implementation, see the [Technical Documentation](#).

For developer integration, see the [SDK Guide](#).

For endpoint reference, see the [API Documentation](#).

FieldFlow · June 2026